Worksheet 4.3 — Bivariate Data and Correlation

Mathematics: Analysis & Approaches SL (IB) · Unit 4: Statistics and Probability

The Big Ideas

Correlation measures linear association; a regression line predicts, but extrapolation is risky.

- r near ± 1: strong linear relationship; near 0: weak/none.
- Interpolation (inside the data range) is more trustworthy than extrapolation (outside it).

Worked Examples

Example 1 — Interpret r

A study finds $r = -0.82$ between hours of TV watched and test scores. Interpret this.

Solution. Strong negative correlation — more TV watched tends to associate with lower scores.

Conclusion: strong negative correlation.

Example 2 — Interpolation

A line $y = 2.1x + 8$ fits data for $x \in [0, 20]$. Predict y at $x = 12$ and classify.

Solution. $y = 2.1(12) + 8 = 25.2 + 8$; $x = 12$ is inside $[0, 20]$.

Conclusion: $y = 33.2$; interpolation.

Example 3 — Extrapolation risk

Using the same line, predict y at $x = 80$ and explain the risk.

Solution. $y = 2.1(80) + 8 = 176$; far outside $[0, 20]$.

Conclusion: $y = 176$ mathematically, but unreliable since it's extrapolated well beyond the data.

Example 4 — Slope interpretation

A line relating monthly rent x (\$100s) to apartment size y (m^2) is $y = 3.5x + 12$. Interpret the slope.

Solution. Each extra \$100 in monthly rent is associated with about 3.5 more m^2 of space, on average.

Conclusion: slope 3.5: more rent associates with more space.

Example 5 — Comparing correlation strength between two data sets

Data set M has $r = 0.61$ between exercise hours and mood score; data set N has $r = 0.88$ between sleep hours and mood score. Which variable shows a stronger linear relationship with mood?

Solution. Compare $|r|$ values: $0.88 > 0.61$.

Conclusion: sleep hours shows the stronger linear relationship with mood.

Example 6 — Predicting and reasoning about reliability near the data's edge

A line $y = 1.8x + 22$ fits data for $x \in [5, 25]$. Predict y at $x = 24$ and comment on reliability.

Solution. $y = 1.8(24) + 22 = 43.2 + 22 = 65.2$.

Conclusion: $x = 24$ is inside $[5, 25]$ (interpolation), so this prediction is reasonably reliable, though close to the data's edge.

Example 7 — Using two data points to build a rough line, then interpreting

A relationship passes near $(2, 10)$ and $(8, 34)$. Find the approximate slope and interpret it.

Solution. $m = \dfrac{34 - 10}{8 - 2} = \dfrac{24}{6} = 4$.

Conclusion: each unit increase in x is associated with about a 4-unit increase in y.

Example 8 — Application — identifying a confounding variable

A study finds a strong positive correlation between the number of firefighters sent to a fire and the amount of damage caused. Explain why this doesn't mean firefighters cause damage.

Solution. Larger fires require more firefighters AND cause more damage — fire size is the confounding variable driving both.

Conclusion: the correlation reflects a shared cause (fire size), not firefighters causing damage.

Example 9 — Extrapolation with a physically impossible result

A line $y = -5x + 120$ models a runner's finishing time (minutes) vs training weeks x, fit for $x \in [0, 15]$. Predict y at $x = 30$ and explain why the result signals extrapolation has failed.

Solution. $y = -5(30) + 120 = -150 + 120 = -30$.

Conclusion: a negative finishing time is physically impossible — a clear sign this extrapolation, far outside $[0, 15]$, has broken down.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Describe the correlation implied by $r = 0.25$.

Question 2

Describe the correlation implied by $r = -0.97$.

Question 3

A line $y = 4x + 15$ fits data for $x \in [2, 18]$; predict y at $x = 10$ and classify (interpolation/extrapolation).

Question 4

Using the same line, predict y at $x = 60$ and explain the risk.

Question 5

A line relates study hours x to score y: $y = 5.2x + 48$; interpret the y-intercept in context.

Question 6

Ice cream sales and sunscreen sales are strongly positively correlated. Explain why this doesn't mean one causes the other.

Question 7 — Challenge

A line $y = -2.4x + 90$ models a car's value (\$1000s) vs its age x (years), fit on data for $x \in [0, 10]$; predict the value at $x = 25$ and explain why this prediction is almost certainly wrong.

Question 8

Data set P has $r = 0.45$ between coffee intake and typing speed; data set Q has $r = -0.79$ between screen brightness and eye strain relief. Which shows the stronger linear relationship?

Question 9

A line $y = 2.2x + 30$ fits data for $x \in [4, 20]$; predict y at $x = 19$ and comment on reliability.

Question 10

A relationship passes near $(3, 15)$ and $(9, 45)$; find the approximate slope and interpret it.

Question 11 — Challenge

A study finds cities with more ice cream shops also have more reported sunburns. Identify the confounding variable and explain.

Question 12 — Challenge

A line $y = -3x + 80$ models a hiker's remaining energy (%) vs distance walked x (km), fit for $x \in [0, 20]$; predict energy at $x = 35$ and explain why the result is physically impossible.

Question 13 — Challenge

A line $y = 6.5x + 40$ relates advertising spend ($100s) to weekly sales; interpret both the slope and the intercept in context.

Question Recap & Answer Key

Question Recap

1. Describe the correlation implied by $r = 0.25$.
2. Describe the correlation implied by $r = -0.97$.
3. A line $y = 4x + 15$ fits data for $x \in [2, 18]$; predict y at $x = 10$ and classify (interpolation/extrapolation).
4. Using the same line, predict y at $x = 60$ and explain the risk.
5. A line relates study hours x to score y: $y = 5.2x+48$; interpret the y-intercept in context.
6. Ice cream sales and sunscreen sales are strongly positively correlated. Explain why this doesn't mean one causes the other.
7. A line $y = -2.4x + 90$ models a car's value ($1000s) vs its age x (years), fit on data for $x \in [0, 10]$; predict the value at $x = 25$ and explain why this prediction is almost certainly wrong.
8. Data set P has $r = 0.45$ between coffee intake and typing speed; data set Q has $r = -0.79$ between screen brightness and eye strain relief. Which shows the stronger linear relationship?
9. A line $y = 2.2x + 30$ fits data for $x \in [4, 20]$; predict y at $x = 19$ and comment on reliability.
10. A relationship passes near $(3, 15)$ and $(9, 45)$; find the approximate slope and interpret it.
11. A study finds cities with more ice cream shops also have more reported sunburns. Identify the confounding variable and explain.
12. A line $y = -3x + 80$ models a hiker's remaining energy (%) vs distance walked x (km), fit for $x \in [0, 20]$; predict energy at $x = 35$ and explain why the result is physically impossible.
13. A line $y = 6.5x+40$ relates advertising spend ($100s) to weekly sales; interpret both the slope and the intercept in context.

✓ Answer Key

1. very weak positive correlation
2. very strong negative correlation
3. $y = 55$; interpolation (10 is inside [2,18])
4. $y = 255$; extrapolation, far outside the fitted range, so unreliable
5. predicted score with 0 study hours is 48 — a baseline from prior knowledge/other factors
6. a confounding variable (hot weather) increases both independently
7. $y = -2.4(25) + 90 = 30$; but a car's value can't go negative and doesn't depreciate linearly forever, so extrapolating this far is unrealistic
8. $|-0.79| > |0.45|$: screen brightness/eye strain shows the stronger relationship
9. $y = 2.2(19) + 30 = 71.8$; $x = 19$ is inside $[4, 20]$ but near the edge, so reasonably reliable but worth caution
10. $m = \frac{45-15}{9-3} = 5$; each unit increase in x associates with about a 5-unit increase in y
11. warmer weather (or tourist season) is the confounding variable — it independently increases both ice cream shop visits and sun exposure/sunburns
12. $y = -3(35) + 80 = -25$; negative remaining energy is impossible, showing the extrapolation (far outside $[0, 20]$) has failed
13. slope 6.5: each extra $100 spent associates with about 6.5 more weekly sales; intercept 40: baseline sales with no advertising spend